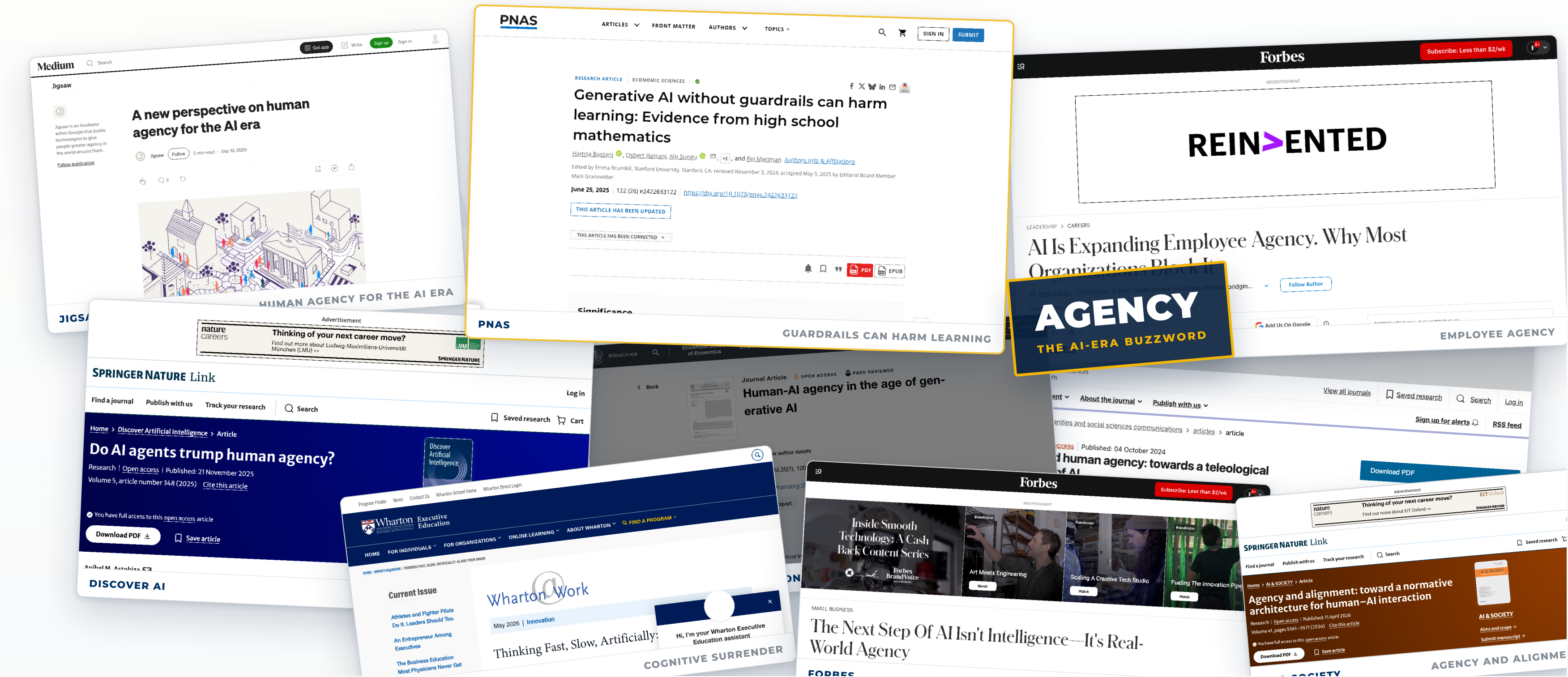


A RESEARCH AND LEARNING LAB INCUBATED AT THE AGENCY FUND

The **Agency** & AI Lab

A²L is a decentralized, virtual, and open lab with two research topics: **design** AI that strengthens human agency in the social sector, and **measure** what AI does to the agency of the people who use it.

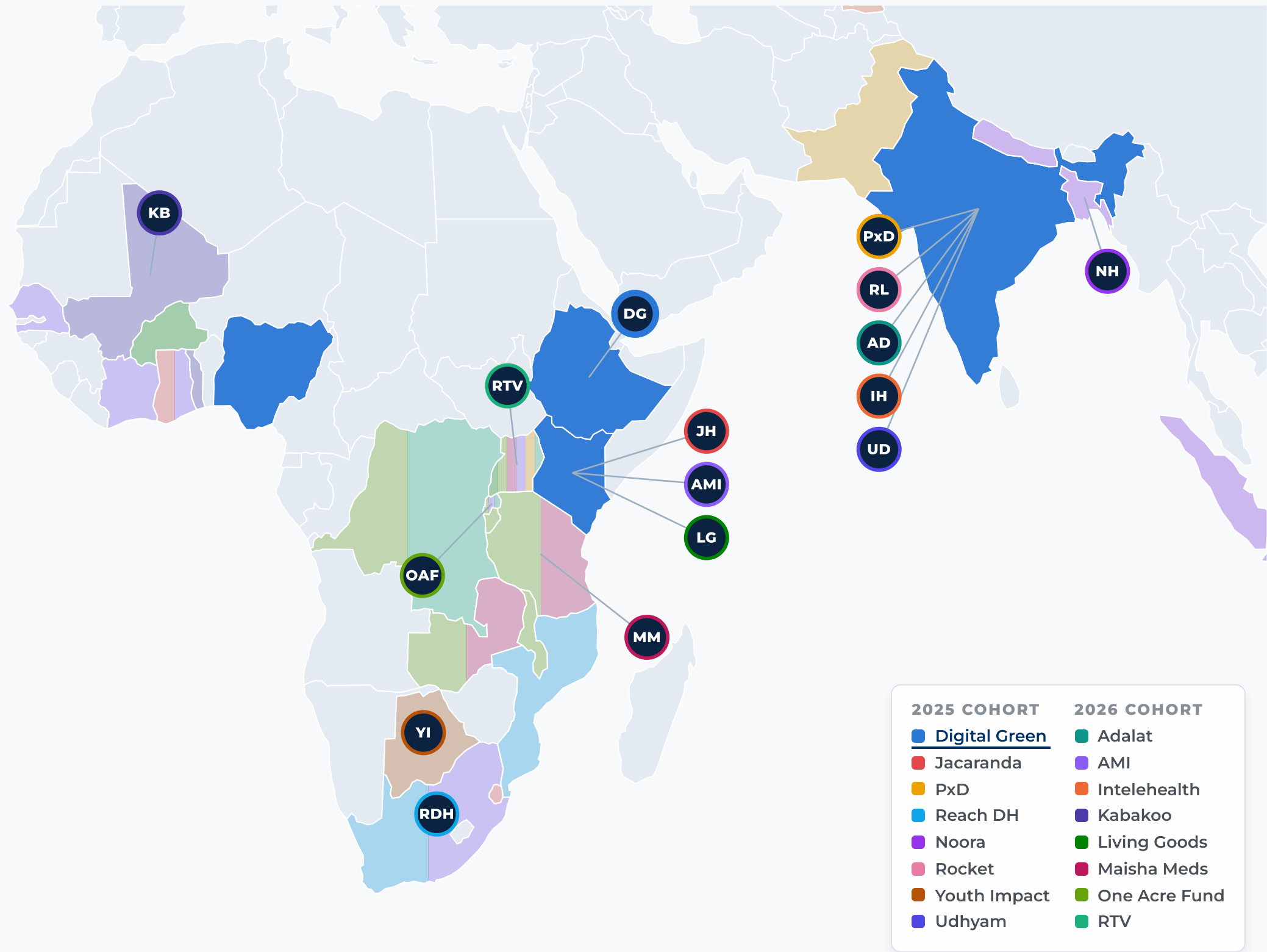
AI can expand agency, and it can hamper it.



Sources: Jigsaw; Bastani et al., PNAS; Shaw & Nave, SSRN; Forbes employee agency; Forbes real-world agency; Forbes human agency; Discover AI; AI & Society; Nature HSSC; Human-AI agency.

We worked with NGOs globally to design and evaluate their AI products through accelerator programs.

INTERACTIVE Click the map, chips, or legend to focus an org · click empty space to clear, or Reset



DG Digital Green 2025 COHORT

FarmerChat

A multilingual AI agronomy advisor for smallholder farmers, using text, voice, and images to deliver localized recommendations.

- L1 · MODEL** Model stack is live across 16 languages, with text, voice, image evals and over 100% improvement over the base model.
- L2 · PRODUCT** 1M+ users have generated 8M+ queries; 30% are monthly active and average 8.7 queries per month.
- L3 · USER** About 70% of active users report applying FarmerChat advice; 80% share advice with peers.
- L4 · IMPACT** Kenya assessment finds adoption of cost-saving practices; income and climate-resilience outcomes are in ongoing RCTs.

AI PRODUCT IN: INDIA · KENYA · NIGERIA · ETHIOPIA · BRAZIL
ORGANIZATION OPERATES IN: INDIA · KENYA · NIGERIA · ETHIOPIA · BRAZIL

Working with these organizations surfaced **two gaps related to human agency** for AI products in the social sector.

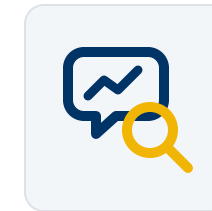


THE DESIGN GAP

AI products are built to optimize service delivery, rather than explicitly to strengthen people's agency.

Despite AI's potential to strengthen human agency, NGO products often treat agency as a byproduct rather than an explicit design objective.

Decades of research in behavioral science and psychologically wise interventions identify agency as a key mechanism for improving development outcomes, yet it is rarely **intentionally built into AI design, testing, or iteration.**



THE MEASUREMENT GAP

AI products are underused as windows into how human agency evolves.

Many products may not generate as many signals of agency as tools like ChatGPT or Claude Code do. Even when these signals exist, they often come from power users, and the field lacks a shared standard or analytical tools for **measuring and interpreting agency** inside human-AI digital traces.

Which health AI chatbot better promotes agency?

INFORMATION DELIVERY

<

A

A2L Health Line

AI assistant · online

Today

hello. my feet r swelling in the evening. i am 7 months pregnant

10:14 ✓

Hello Priya. Mild evening swelling is common in month 7. It helps to:
1. Rest with your feet raised.
2. Drink water through the day.
3. Avoid standing for long.
Go to the clinic if swelling is sudden or comes with headache or blurred vision.

10:15

ok. can i still do house work

10:22 ✓

Light housework is fine. Avoid heavy lifting and take short breaks.

10:23

ok thank you

10:26 ✓

Message

AGENCY DEVELOPMENT

<

A

A2L Health Line

AI assistant · online

Today

hello. my feet r swelling in the evening. i am 7 months pregnant

10:14 ✓

Hello Priya. Evening swelling is common in month 7, and resting with feet raised helps. Have you managed your ANC check-up this month?

10:15

not yet. clinic is far and my husband takes the scooter for work

10:21 ✓

That is a real constraint. The 7th-month check matters — they will look at this swelling and your BP. Which day is he home?

10:22

thursday, home by 2

10:24 ✓

Then a plan: Thursday at 3 he drops you at the clinic. Carry your mother-child card. Shall I remind you at 1?

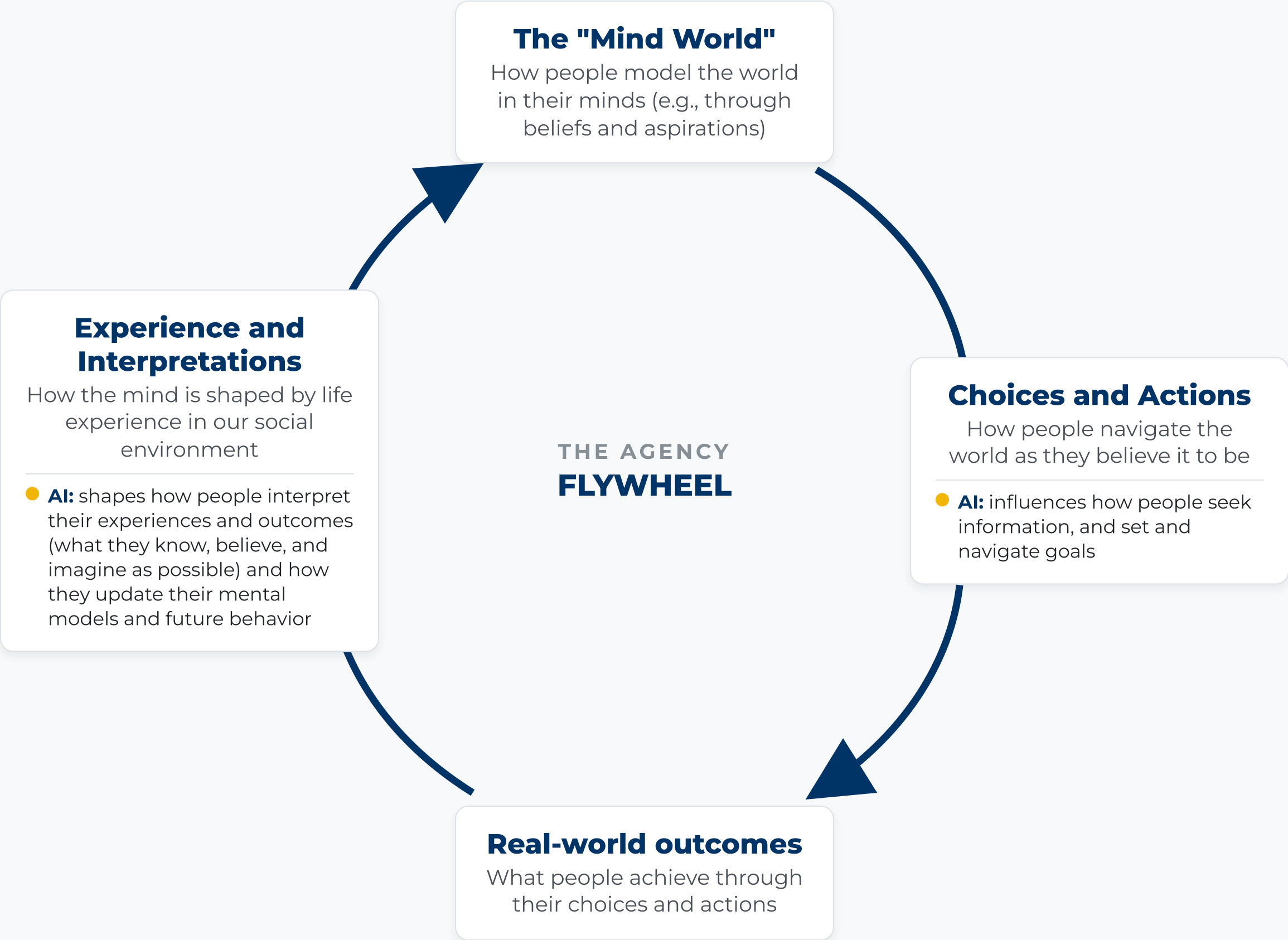
10:25

yes pls. thursday 3 pm i will go

10:27 ✓

Message

Our worldview of agency in a flywheel



The loop can turn in either a good or a bad way.

When it turns well, capable choices bring results that promote a positive cycle. When it turns badly, failed choices result in a negative cycle.

AI products can influence how the flywheel turns.

An AI tool can push any of the four elements in either direction, while generating lots of digital behavioral traces.

That's exactly why we need to design AI for human agency and be able to concretely measure agency.

PART 2 · REPEATABLE MOTIONS

**We created repeatable motions
based on our work with the
NGOs.**

Open-source tools, papers, and AI skills that connects psychology with AI

A research library of agency interventions

A Google NotebookLM as an interactive agency intervention learning assistant

notebooklm.google.com · Agency-Based Intervention Assistant

Agency Intervention Learning Assistant by The Agency Fund

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Select all

Ecological Systems Theory

Implementation Science

Intervention Mapping

Planned Behavior Theory

Psychological Interventions

Self-Determination Theory

Social Cognitive Theory

Social Policy Research

Chat

Agency Intervention Learning Assistant by The Agency Fund

29 sources · Jan 3, 2026

Icek Ajzen's 1991 paper introduces the **Theory of Planned Behavior**, a psychological framework designed to predict and explain human actions. This model expands upon the earlier **Theory of Reasoned Action** by adding the concept of **perceived behavioral control**, which accounts for activities that individuals cannot fully influence through will alone. The author argues that **intentions**—driven by personal attitudes, social norms, and the perceived ease of an action—serve as the primary motivators for behavior. While empirical evidence largely supports the theory, the source also explores **unresolved issues** such as the role of past habits and the impact of moral obligations. Ultimately, the text demonstrates that combining **motivation and perceived ability** provides a highly accurate method for understanding how complex social behaviors are formed.

Save to note

Sunday, January 4

what is agency

Agency is the capacity to intentionally make things happen by one's actions; it involves the endowments, belief systems, and self-regulatory capabilities through which people exercise influence over the nature and quality of their lives. It is the essence of humanness.

Gemini Notebook just got smarter. Try asking to create documents and spreadsheets.

Ask a question or create something

29 sources

Studio

Audio Overview

Slide Deck

Video Overview

Mind Map

Reports

Flashcards

Quiz

Infographic

Data Table

New Note

Architectures of Human Agency and Intervention...

How to Use This Notebook

[Quiz] Q&A on the Core Theoretical Ideas

[Flashcards] Core Theoretical Ideas

[Table] Summary of Theories and Applications

[Audio] Behavioral Science Theories f...

[Flashcards] Pra...

Add note

Gemini Notebook can be inaccurate; please double check its responses.

A research library of agency measurement tools

A Google NotebookLM for finding, comparing, and adapting agency measurement tools, co-developed with Google Jigsaw

notebooklm.google.com · Agency Measurement Tool Library

Measuring Agency in a Digital Context @ The Agency Fund & Jigsaw

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+ Add sources

Search the web for new sources

Web

Fast Research

Select all

Child Development

Education

Empowerment

Healthcare

Identity

Methods

Motivation

Other

Political Science

Self-Efficacy

Social Behavior

Technology

Theories of Agency

GenAI Evaluation Playbook in the Development Se...

Chat

Measuring Agency in a Digital Context @ The Agency Fund & Jigsaw

184 sources · Jun 21, 2026

Customize

Welcome to this NotebookLM collection on **measuring agency in digital contexts**.

This notebook brings together theoretical papers, useful scales, and other resources on how agency can be understood, measured, and supported in digital environments. It was created as companion material for a blog drafted by [The Agency Fund](#) and [Jigsaw](#).

You can use this notebook to explore the research, compare measurement approaches, and brainstorm ideas for measuring agency in digital contexts. Please note that tags shown in the left sidebar are generated by NotebookLM.

Like any AI tool, NotebookLM can be helpful for thinking with documents, but it can also make mistakes. Please use it at your own discretion and refer back to the original sources when accuracy matters.

We hope this notebook serves as a useful starting point for research, discussion, and new ideas around measuring agency. We also welcome your feedback through [this form](#).

Creator Notes

Save to note

Gemini Notebook just got smarter. Try asking to analyze your data and create charts.

Ask a question or create something

184 sources

Studio

Audio Overview

Slide Deck

Video Overview

Mind Map

Reports

Flashcards

Quiz

Infographic

Data Table

Taxonomy of Agency and Agency-Related...

184 sources · 48d ago

Agency Map

184 sources · 48d ago

Agency Flashcards

184 sources · 48d ago

How AI erodes your agency

21:21 · Deep Dive · 182 sources · 56d ago

Add note

Gemini Notebook can be inaccurate: please double check its responses.

A review paper that connects AI eval with cultural defaults

Interdisciplinary paper from the accelerator's AI eval working group, PNAS Nexus (under review)

PNAS Nexus, 2026, , 1–13
DOI added during production
Published: Date added during production
Paper



Evaluating AI Across Cultures: A Multilevel Framework for Generative AI in Global Development

Ryan L. Boyd^{1,*}, Zezhen Wu^{2,3}, Ellen C. Reinhart⁴, Robert On², David M. Markowitz⁴, Aman Dalmia², Vasudha Varadarajan⁵, Han Sheng Chia⁶, Weixi Wang⁷, Sunayana Sitaram⁸, Shi Feng⁹, Markus Goldstein⁶ and Hazel R. Markus^{10,11}

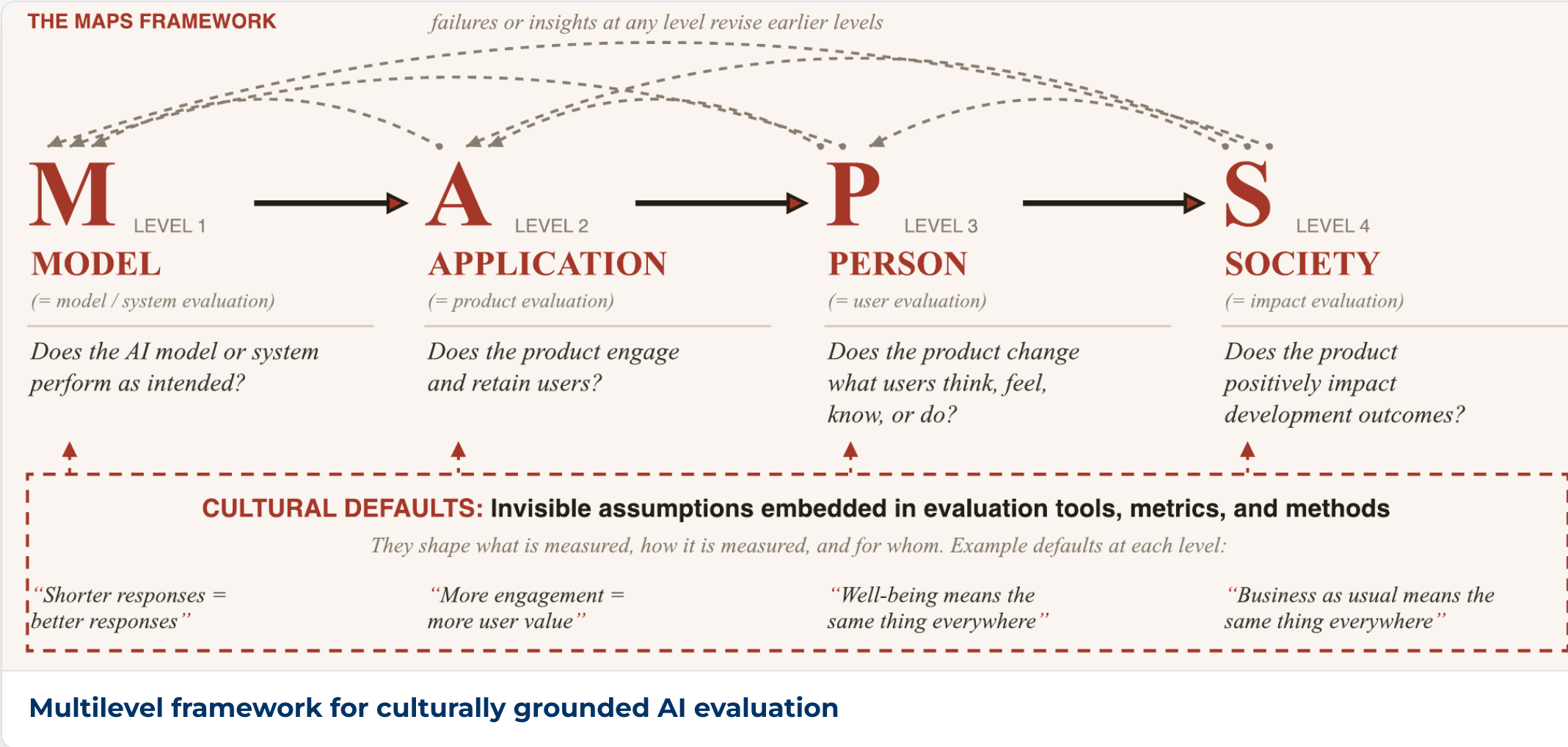
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²The Agency Fund, San Francisco, CA, USA
³The Institute of Human Development and Social Change, New York University, New York City, NY, USA
⁴Department of Communication, Michigan State University, East Lansing, MI, USA
⁵Language Technologies Institute, Carnegie Mellon University, Pittsburgh, PA, USA
⁶Center for Global Development, Washington, DC, USA
⁷Department of Psychology, The University of Texas at Dallas, Richardson, TX, USA
⁸Microsoft Research, Bangalore, India
⁹Department of Computer Science, George Washington University, Washington, DC, USA
¹⁰Department of Psychology; Stanford SPARQ, Stanford University, Stanford, CA, USA
¹¹To whom correspondence should be addressed: [TODO: update with vanderbilt address when assigned] ryan@ryanboyd.io

Abstract

Generative AI tools are being deployed at scale across low- and middle-income countries as health chatbots, agricultural advisors, educational tutors, and beyond, yet the frameworks used to evaluate them were developed within Western, English-dominant, commercially oriented contexts. This cultural mismatch creates a structural problem: evaluation tools that function well in high-income settings can produce systematically flawed conclusions when applied to diverse global populations. Drawing on insights from a multidisciplinary working group spanning computer science, economics, behavioral science, public health, and linguistics, we introduce a four-level evaluation framework MAPS (the Model, its Application, the Person who uses it, and the Society it affects), designed to make explicit how culture shapes AI evaluation at every level. We argue that culture operates not as a moderating variable to be statistically controlled but as a set of defaults: invisible assumptions embedded in benchmarks, engagement metrics, psychological measures, and outcome assessments. When these defaults do not match the deployment context, evaluation produces artifacts rather than evidence. Grounding the framework in case studies from AI deployments across Sub-Saharan Africa and South Asia, we show how culturally responsive evaluation differs from standard practice at each level and propose a research agenda for this emerging field.

Keywords artificial intelligence, culture, evaluation, global development, language diversity, WEIRD

Paper draft



Agency and behavioral science skills that **ride along** inside everyday AI work

A collaboration between The Agency Fund and Irrational Labs, with Joe Speed's Claude Skills open source at github.com/agency-fund/taf-skills

joe-speed.github.io/behavioural-skills

Behavioural Skills

CatalogueWorkflow BuilderPropose a Skill

Search

Star on GitHub 0

A curated, chainable library of atomic AI skills for behavioural science — each one narrow enough to describe without the word "and," built so one skill's output can feed the next.

Search skills by name or description...

CATEGORY

AllBehaviour DefinitionDiagnosisIntervention DesignMeasurementResearch MethodsSegmentation & PersonasRisk & Pre-MortemCultural ContextEvidence ScopingResearcher CalibrationContext MappingResearch Transparency

RESEARCH STAGE

AllPrepareDefineDiagnoseDesignTestMeasureReport

COM-B Barrier Decomposer

DiagnosisDiagnose

Use when one specific, already-defined target behaviour exists (a clear who + what + when, not a goal or outcome) and the question on the table is "why isn't this happening yet?"

Context & Audience Mapper

Context MappingPrepare

Use when a program goal names its target population only in the abstract identity terms a funder or stakeholder would use (e.g. "smallholder farmers", "gig workers", "first-time mothers") with no detail on the actual...

Install

Select skills from the list to build a combined install command.

INSTALL COMMAND

Copy, paste, run. Updates as you change platform or selection.

Platform

macOS / Linux

Selected: 0

BASH

Copy

```
curl -sSL https://raw.githubusercontent.com/Joe-Speed/behavioural-skills/main/scripts/install.sh | bash -s -- --target . --name <select-skills>
```

Advanced

Behavioral-science skills for applied AI workflows

Claude Skills · Agency Measurement

The Agency Fund

Afternoon, Zezhen

/ask-agency-measurement

what is agency and why should I measure it?

+Sonnet 4.6 High

🎤

⌂

Agency-measurement skills for research and evaluation

PART 3 · RESEARCH: DESIGNING AI FOR AGENCY

Design the agency in — don't bolt it on.

Field research inside real NGO products, asking what AI design does to users' agency — with the psychology brought in from the start.

Evaluating a live copilot to align AI design with the agency of the humans in the loop

Research · a mixed-methods case study with Noora Health and the Stanford Center for Health Education

THE SETTING

Noora Health's **Mobile Care Companion** answers families' health questions over WhatsApp. Since November 2025, a **GenAI copilot** reads incoming queries and drafts language-matched replies for Medical Support Executives (MSEs), who review, edit, and send every message. The copilot never faces patients: its effects run through MSE judgment.

240_k

queries per month across India

80_k

users reached monthly

15_k

health facilities

6

languages, Hindi to Odia

THE STUDY · CONCURRENT MIXED METHODS

Copilot access
9 MSEs

randomized 1:1,
by language

No copilot
9 MSEs

- RQ1** How do MSEs **experience** the copilot: attitudes, trust, agency, everyday practice? 18 interviews across both arms, plus every edit to an AI draft read as a behavioral trace of where the tool meets or falls short of MSEs' needs.
- RQ2** Do **conversations** measurably differ? Expert rubrics for clarity, medical safety, empathy, and clear next steps, scaled with a validated LLM judge across the full corpus in English and Indic languages.
- Then** Findings **triangulated**: multilevel models compare the two groups, interpreted alongside what MSEs say in interviews.

The point of the evaluation is to align copilot design with MSEs' agency: findings feed the next iteration, so the tool strengthens frontline judgment rather than replacing it.

Three papers, one call: put psychology inside AI design and evaluation

Research · the agenda the lab builds on

DESIGN

THE CORE ARGUMENT

Chatbots could democratize personalized support, or deepen isolation and risk. The difference is whether **psychological theory and methods are embedded** in design, evaluation, and monitoring, with concrete roles for psychologists across that whole pipeline.

FLIP BACK

ALIGNMENT

THE CORE ARGUMENT

AI should be understood as an **extension of human purposiveness**, grounded in practical autonomy: the human capacity to act for reasons and take responsibility. Alignment then runs through a normative interface connecting machine behavior to human norms, with humans remaining the ultimate bearers of responsibility.

FLIP BACK

EVALUATION

THE CORE ARGUMENT

Benchmarks reward accuracy, reasoning, and compliance, but say little about whether a system supports **user cognition, emotional interpretation, and behavioral decision-making** in ways fit for the user and context. The authors propose testing this with scenario-based probes and structured human and model-assisted evaluation.

FLIP BACK

Most NGO AI products in the Global South are built without this expertise in the room. That is the design gap, and **the lab's opening.**

Sources: [Psychologists must be involved in building conversational AI chatbots](#) (Nature Reviews Psychology, 2026) · [Agency and alignment: toward a normative architecture for human-AI interaction](#) (AI & Society, 2026) · [Psychological competence as a missing dimension in AI evaluation](#) (arXiv, 2026).

PART 4 · MEASURING AGENCY

You cannot design for what you **cannot see.**

Agency measurement is interesting, necessary, and complex. What follows is the approach we teach and use, and the flagship study that puts it to work at scale.

We wrote the measurement playbook down, with Google Jigsaw

Measuring agency · a co-published practitioner's guide · theagencyfund.substack.com



Measuring Agency in a Digital Context

A practitioner's guide by **Zezhen Wu** (The Agency Fund) with **Beth Goldberg and Rachel Xu** (Google Jigsaw), published July 2026.

The method, in three moves

Define what agency means in your context, **choose** measures across on-platform and off-platform data, then **test and validate** them iteratively.

Ships with tools

An open Claude Skill for brainstorming measurement approaches, and a curated NotebookLM of the underlying research. Free for any team.

The lab's measurement thinking is public and reusable, and **the next slides walk its core moves.**

What you need to measure also depends on the forms of agency and how agency manifests in your product.

Hypothetical example: an AI mentor helping learners develop a viable business idea

Agency with respect to the end goals

Can users make progress toward outcomes they value, whether through, around, or beyond the tool?

Internally experienced agency

People's thoughts and feelings

ON-PLATFORM DATA • IN-APP MICRO-SURVEY
Short in-app surveys probing users' general sense of self-efficacy or perceived control over their ultimate goal over time (e.g., "How confident is the user about developing a successful business idea? How much control do they feel over reaching their ultimate goal?")



OFF-PLATFORM DATA • PHONE OR IN-PERSON SURVEY
Longer phone/in-person surveys using validated agency measures that systematically probe users' general sense of self-efficacy or perceived control over their ultimate goal over time.



FLIP BACK

Externally expressed agency

People's choices and actions

ON-PLATFORM DATA • CONVERSATION LOGS
Longitudinal user signals of agency in AI conversation logs, especially those related to setting and navigating longer-term goals or demonstrating resilience despite difficulties.



OFF-PLATFORM DATA • RUBRIC-RATED PITCHES
Longer phone/in-person surveys tracking how users develop their business ideas; rubric-rated observations of ideas and pitches; or classroom observation of teamwork, ideally linked to downstream outcomes like business earnings.



FLIP BACK

Agency with respect to the AI tool

Can users understand, question, adapt, and control the tool, deciding when and how to rely on it?

ON-PLATFORM DATA • IN-APP MICRO-SURVEY
Short in-app surveys probing whether users felt able to understand, question, or adapt an AI mentor's advice.



OFF-PLATFORM DATA • VALIDATED SCALES
Longer phone/in-person surveys using validated agency measures that systematically probe users' trust in IA and perceived dependence versus empowerment when using a tool.



FLIP BACK

ON-PLATFORM DATA • DIGITAL TRACES
Digital traces (e.g., interaction patterns or conversation logs) showing whether users understood, questioned, or adapted an AI mentor's advice.



OFF-PLATFORM DATA • OBSERVATION AND INTERVIEWS
Observation or interview data (e.g., usability studies) on tool usage for developing business ideas.

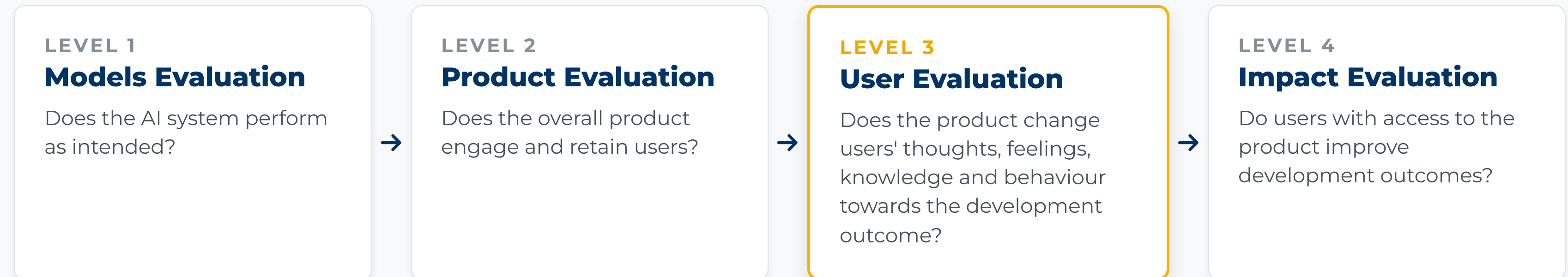


FLIP BACK

 **Mind the social setting:** norms and relationships can also shape these questions; interpret the answers in your users' contexts.

Agency measurement sits at **Level 3**

The four-level evaluation framework · eval.playbook.org.ai



Nobody can run an RCT on every product change. **Validated agency measures at Level 3** move when the product changes, and they speak to what Level 4 cares about. That also allows teams to anchor their A/B tests on intermediate measures that are psychologically and behaviorally meaningful and relatively easy to collect.

Teams should specify what agency means for **their specific users and contexts**

1

Define agency for your local problem

What kind of agency is the tool supposed to support? **For whom? To what end?** Anchor the definition in your program's theory of change.



2

Measure with the right tools, on the right data, at the right moment

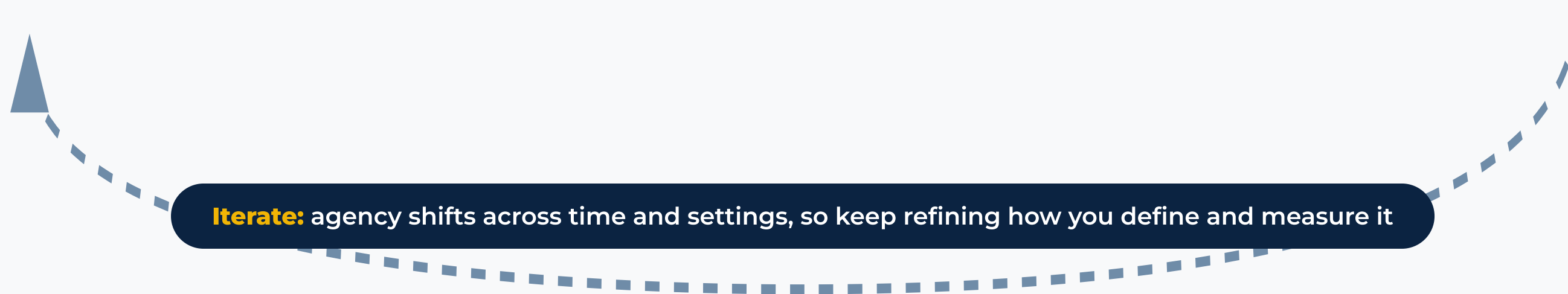
Combine **on-platform** signals, such as in-app surveys and digital traces, with **off-platform** surveys, interviews, observations, and records.



3

Test and validate your measurement tools iteratively

Test for evidence of **reliability and validity** for your context.



The flagship study: measure agency **where it already shows**

Research · with Hamsa Bastani and Osbert Bastani, University of Pennsylvania · the team behind the PNAS 2025 guardrails study

WHY NATURALISTIC DATA

Digital platforms already produce the evidence:

→ **conversations, decisions**, and, where partners can link them, **real outcomes**.

→ Surveys, today's default, **interrupt people**, carry self-report bias, and sample thin snapshots in time.

→ Traces are continuous: agency can be observed **as it develops**, unobtrusively, over months instead of moments.

HOW THE METHOD WORKS

1 Transcripts
Text, voice, and video from real platform use.

2 LLM judge
Reads them against a structured rubric grounded in Social Cognitive Theory: **intentionality, forethought, self-reactiveness, self-reflectiveness**.

3 Calibrated scores
Agency scores that carry uncertainty and say so when the evidence is missing.

4 Validation
Expert labels and lightweight micro-surveys check the judge where it matters most.

The goal is a **shared, validated instrument** for reading agency in the wild, built and tested with six NGO partners.

Six partners, six ways to read agency from real products

Research · what we might learn with each organization · health, education, and livelihoods across five countries

JH Jacaranda Health HEALTH · KENYA

PROMPTS, maternal-health SMS with AI triage; 3.8M+ mothers reached.

Read how mothers navigate care pathways, link those signals to actual care visits, and design proactive nudges for mothers who do not yet know what to ask.

DG Digital Green LIVELIHOODS · KENYA · ETHIOPIA · INDIA

FarmerChat agronomy advisor; 1.3M farmers.

Find agency signals among power users, track how they shift from version 1 to version 2, and test whether they predict real-world adoption.

KA Kabakoo Academies LIVELIHOODS · MALI

WhatsApp AI mentor with video reflections, often in Bambara.

Mine reflections for agency signals and connect them to income and community standing in the years ahead, under strict privacy protections.

NE Nova Escola EDUCATION · BRAZIL

WhatsApp lesson planner on the national curriculum; 150K teachers.

Test whether co-creating plans in conversation builds lasting teaching practice or only saves time, by varying copilot features.

TAP The Apprentice Project EDUCATION · INDIA

TAP Buddy project-based learning, plus mother-tongue voice AI.

Measure creativity and 21st-century skills as agency signals, and learn how AI feedback changes what students attempt next.

U Udhyam Learning Foundation EDUCATION · INDIA

AI help bot inside a real-business entrepreneurship program.

Infer grit, motivation, and self-efficacy from chats and submissions, and feed the findings back into the program and the bot.

Every partner runs a live AI product with real users and real stakes. Together they give the study conversations in **text, voice, and video** across languages and cultures.

Let's build AI that
strengthens human agency,
and **prove that it does.**